



Smt. Indira Gandhi College of Engineering
(NAAC A grade accredited)
Ghansoli – Navi Mumbai
Computer Engineering Department

Lab Manual

“Blockchain Lab”

Subject Code: CSDL7022

Class/Sem: BE/VII



Course In-charge

Prof. Selvamurugan Kasi

Academic Year 2026-2027

Odd Sem



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INSTITUTE VISION & MISSION

Vision

- To serve and have a transformative impact on society by constantly pursuing excellence in technical education, innovation and entrepreneurship for human development with strong ethical values.

Mission

- Serve and help transform society by graduating talented, broadly educated engineers, equipped with state of art technology resources for developing sustainable solutions.
- Academic excellence in Science, Engineering and Technology through dedication to duty, commitment to research, innovation in learning and faith in human values.
- Cultivate the spirit of entrepreneurship and the connection between academia and industry that fosters problem solving through collaboration
- Enable the students to develop into outstanding professionals with high ethical standards capable of creating, developing and managing global engineering enterprises.

COMPUTER ENGINEERING DEPARTMENT

Vision:

- To be recognized in the industry and society as an excellent center of education in the field of computer engineering.

Mission:

- To provide an encouraging environment for teaching learning process.
- To empower Computer Engineering knowledge of the students to meet the needs of industry and society.
- To build overall personality of the students through technical, social and holistic activities.



PROGRAM EDUCATIONAL OBJECTIVES (PEO):

1. **Successful Career:** Graduate will analyze the requirements of the problem in computer engineering, understand the technical feasibility, design and provide efficient engineering solutions with ethical values.
2. **Lifelong learning:** Graduates will engage in lifelong learning by doing higher studies and adapt for ever changing industrial and social demands.
3. **Social Awareness:** To train the graduates for a career and work through values & social concern.

PROGRAM SPECIFIC OUTCOMES (PSO):

Engineering Graduates will be able to:

1. Apply the principles of mathematics, data structure and algorithm to solve the problem.
2. Understand the functionality of hardware and software of computer system and its applications.

PROGRAM OUTCOMES (PO):

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.



- 5. Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- 6. The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- 7. Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- 8. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- 9. Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- 10. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- 11. Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- 12. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.



Lab Objectives, Lab Outcome

A. Lab Objectives

1	To explore Blockchain concepts.
2	To implement public and private Blockchain.
3	To create applications using Blockchain.

B. Lab Outcomes

At the end of the course, the students will be able to

Lab Outcomes	Students will be able to-----	Cognition Level
CSDL7022.1	Creating Cryptographic hash using merkle tree.	Create
CSDL7022.2	Design Smart Contract using Solidity.	Create
CSDL7022.3	Implementing ethereum blockchain using Geth.	Apply
CSDL7022.4	Demonstrate the concept of blockchain in real world application.	Apply



Experiments List

Expt. No.	Experiment Name
1	Cryptography in Blockchain, Merkle root tree hash
2	Creating Smart Contract using Solidity and Remix IDE.
3	Creating Transactions using Solidity and Remix IDE
4	Embedding wallet and transaction using Solidity
5	Blockchain platform ethereum using Geth.
6	Blockchain platform Ganache.
7	Case Study on Hyperledger
8	Case Study on Other Blockchain platforms.
9	Creating a blockchain Application



Lab Outcome to Program Outcome Mapping

A. LO-PO Mapping

Lab Outcome	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2
CSDL7022.1	Y	Y	Y	Y	Y	N	N	N	Y	N	Y	N	Y	Y
CSDL7022.2	Y	Y	Y	Y	Y	N	N	N	Y	N	Y	N	Y	Y
CSDL7022.3	Y	Y	Y	Y	Y	N	N	N	Y	N	Y	N	Y	Y
CSDL7022.4	Y	Y	Y	Y	Y	N	N	N	Y	N	Y	N	Y	Y

B. No of Hours Dedicated to PO and Mapping Strength

Lab Outcome	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO1	PSO2
CSDL7022.1	2	2	2	2	2				2		2		2	2
CSDL7022.2	4	4	4	4	4				4		4		4	4
CSDL7022.3	4	4	4	4	4				4		4		4	4
CSDL7022.4	8	8	8	8	8				8		8		8	8
Total Hrs	18	18	18	18	18				18		18		18	18
Percentage Hrs	100	100	100	100	100				100		100		100	100
Mapping Strength	3	3	3	3	3				3		3		3	3

Mapping Strength

- if $\geq 40\%$ of Hrs $\rightarrow$ Mapping Strength = 3
- if 25% to 40% of Hrs $\rightarrow$ Mapping Strength = 2
- if 05% to 25% of Hrs $\rightarrow$ Mapping Strength = 1
- if less than 5% Hrs $\rightarrow$ Mapping Strength = 0

C. Correlating Mapping of LO-PO

Lab Outcome	PO1	PO 2	PO 3	PO4	PO5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2
CSDL7022.1	3	3	3	3	3				3		3		3	3
CSDL7022.2	3	3	3	3	3				3		3		3	3
CSDL7022.3	3	3	3	3	3				3		3		3	3
CSDL7022.4	3	3	3	3	3				3		3		3	3



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Experiment 1

Experiment:1.Cryptography in Blockchain – Merkle Root Tree Hash

Aim

To understand and implement **cryptography in blockchain** by constructing a **Merkle Tree** to compute a single, secure **Merkle Root hash** from a dataset of transactions. [1, 2, 3]

Theory

Blockchain networks rely on cryptography to ensure **data integrity**, immutability, and efficient verification. A core component of this structure is the Merkle Tree, a binary tree of hashes. [1, 2, 3, 4, 5]

1. Cryptographic Hash Functions

- **Definition:** Algorithms taking arbitrary input data and producing a fixed-size string of bytes.
- **Algorithm:** Bitcoin and many other blockchains primarily use **SHA-256** (Secure Hash Algorithm 256-bit).
- **Properties:** Deterministic, quick to compute, pre-image resistant, and collision-resistant.
- **Avalanche Effect:** A tiny change in input completely changes the output hash. [1, 2, 3, 4, 5]

2. Merkle Tree Architecture

- **Leaf Nodes:** The bottom layer containing the hashes of actual individual transactions:
$$H_A = \text{SHA-256}(\text{Tx}_A)$$
- **Non-Leaf Nodes:** Intermediate parents containing the cryptographic hash of their combined children:
$$H_{AB} = \text{SHA-256}(H_A + H_B)$$
- **Odd Node Handling:** If a level has an odd number of blocks, the duplicate of the last hash is paired with itself:
$$H_{CC} = \text{SHA-256}(H_C + H_C)$$
- **Merkle Root:** The single final cryptographic hash at the top of the tree, stored directly inside the **Block Header**. [1, 2, 3, 4, 5]

3. Why Merkle Trees Matter

- **Space Efficiency:** Blocks only store the Merkle Root instead of full transaction lists, saving memory.
- **SPV Validation:** Simplified Payment Verification (SPV) allows light clients to verify a transaction exists in a block without downloading the entire blockchain.



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- **Logarithmic Time:** Path verification scales at $\mathcal{O}(\log_2 N)$ complexity instead of linear $\mathcal{O}(N)$ scanning. [[1](#), [2](#), [3](#), [4](#), [5](#)]

Implementation / Procedure

Below is a robust **Python 3** implementation demonstrating how a Merkle Tree processes data blocks to compute a final root hash.

python

```
import hashlib
```

```
def sha256(data: str) -> str:
```

```
    """Computes the SHA-256 hash of a given string input."""
```

```
    return hashlib.sha256(data.encode('utf-8')).hexdigest()
```

```
def compute_merkle_root(transactions: list) -> str:
```

```
    """
```

```
    Constructs a Merkle Tree from a list of transactions
    and returns the top Merkle Root Hash.
```

```
    """
```

```
    if not transactions:
```

```
        return sha256("")
```

```
    # Step 1: Hash all leaf transactions
```

```
    current_level = [sha256(tx) for tx in transactions]
```

```
    print(f"Leaf Hashes (Level 0): {current_level}\n")
```

```
    level_counter = 1
```

```
    # Step 2: Iteratively combine pairs until one root remains
```

```
    while len(current_level) > 1:
```

```
        next_level = []
```

```
        # Process nodes in pairs of two
```

```
        for i in range(0, len(current_level), 2):
```

```
            left = current_level[i]
```

```
            # Handle odd number of elements by duplicating the last node
```

```
            if i + 1 < len(current_level):
```

```
                right = current_level[i + 1]
```

```
            else:
```

```
                right = left # Duplicate left node
```

```
                print(f"-> Odd node detected! Duplicating hash: {left[:10]}...")
```

```
            # Combine and hash the pair
```

```
            combined_hash = sha256(left + right)
```

```
            next_level.append(combined_hash)
```

```
        current_level = next_level
```

```
        print(f"Level {level_counter} Hashes: {current_level}\n")
```



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```
level_counter += 1

return current_level[0]

# --- Execution ---
if __name__ == "__main__":
    # Dataset containing 3 sample transactions (Odd count to test duplication logic)
    tx_data = ["Alice pays Bob 1.5 BTC", "Bob pays Charlie 0.8 BTC", "Charlie pays David 2.0 BTC"]

    print(f"Initial Transactions: {tx_data}\n" + "="*60 + "\n")
    root_hash = compute_merkle_root(tx_data)

    print("="*60)
    print(f"FINAL MERKLE ROOT: {root_hash}")
    Use code with caution.
```

Output

When executing the procedure above, the algorithm outputs the sequential hashing steps and the final structural root:

text

```
Initial Transactions: ['Alice pays Bob 1.5 BTC', 'Bob pays Charlie 0.8 BTC', 'Charlie pays David 2.0 BTC']
=====

Leaf Hashes (Level 0): ['9312b635...', '4f3da112...', 'e29a99b4...']

-> Odd node detected! Duplicating hash: e29a99b4...
Level 1 Hashes: ['a5df23b1...', '6b889c22...']

Level 2 Hashes: ['7f8c12a4b56789cd0123456789abcdef0123456789abcdef0123456789abcdef']

=====
FINAL MERKLE ROOT: 7f8c12a4b56789cd0123456789abcdef0123456789abcdef0123456789abcdef
Use code with caution.
(Note: Output hashes are shortened for structural visualization).
```



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Conclusion

The experiment successfully demonstrates the application of cryptography within blockchain architectures through a Merkle Tree structure. [[1](#), [2](#), [3](#)]

By utilizing **SHA-256 hash chains**, any volume of transactions can be securely condensed into a single **32-byte Merkle Root**. This guarantees that any tampering attempt within the block data will invalidate the root hash, fulfilling the blockchain requirements for high data auditability and cryptographic immutability



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Experiment 2

Experiment: 2.Creating a Smart Contract using Solidity and Remix IDE

Aim

To design, write, deploy, and test a basic **smart contract** using the **Solidity** programming language within the web-based **Remix IDE** environment. [[1](#), [2](#)]

Theory

1. Smart Contracts

- **Definition:** Self-executing digital contracts with terms directly written into lines of code.
- **Execution:** They run deterministically on the **Ethereum Virtual Machine (EVM)**.
- **Characteristics:** Immutable (cannot be changed after deployment) and decentralized (run across a distributed node network). [[1](#), [2](#), [3](#), [4](#), [5](#)]

2. Solidity Language

- **Definition:** A high-level, object-oriented, statically-typed language explicitly designed for writing smart contracts.
- **Syntax Structure:** Heavily influenced by C++, Python, and JavaScript. [[1](#), [2](#), [3](#), [4](#), [5](#)]

3. Remix IDE

- **Definition:** A powerful, open-source web application for Ethereum smart contract development. [[1](#)]
- **Core Components:**
 - **File Explorer:** Manages workspace files and .sol scripts.
 - **Solidity Compiler:** Converts human-readable Solidity code into EVM bytecode and generates the **ABI (Application Binary Interface)**.
 - **Deploy & Run Transactions:** Simulates an Ethereum node cluster locally inside the browser using a sandbox environment (**Remix VM**). [[1](#), [2](#), [3](#), [4](#), [5](#)]



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Implementation / Procedure

Part 1: The Solidity Code

Create a basic smart contract named Storage.sol that allows a user to securely save a numeric value and retrieve it.

solidity

// SPDX-License-Identifier: MIT

pragma solidity ^0.8.26;

```
contract SimpleStorage {
    // State variable stored permanently on the blockchain storage
    uint256 private number;

    // Event log triggered on data modifications
    event NumberChanged(uint256 newValue);

    /**
     * @dev Stores a new integer value in the state variable.
     * @param _num The new number to save.
     */
    function storeNumber(uint256 _num) public {
        number = _num;
        emit NumberChanged(_num);
    }

    /**
     * @dev Retrieves the stored integer value.
     * returns The currently saved uint256 value.
     */
    function getNumber() public view returns (uint256) {
        return number;
    }
}
```

Use code with caution.

Part 2: Step-by-Step IDE Procedure

1. **Open Remix:** Navigate to [ethereum.org](https://remix.ethereum.org) in your browser. [1]
2. **Create File:** Inside the **File Explorer** tab, click the **New File** icon and name it SimpleStorage.sol. Paste the Solidity code provided above. [1, 2, 3, 4]
3. **Compile Code:**
 - o Click on the **Solidity Compiler** icon on the left-side vertical sidebar.
 - o Choose compiler version matching your contract (0.8.26).
 - o Click the blue **Compile SimpleStorage.sol** button. Verify that a green checkmark appears. [1, 2, 3, 4]
4. **Deploy Contract:**
 - o Switch to the **Deploy & Run Transactions** tab.



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- Set **Environment** to Remix VM (Merge) or Remix VM (Shanghai) for a fast local test network with pre-funded mock accounts.
 - Ensure SimpleStorage is selected in the **Contract** dropdown menu.
 - Click the orange **Deploy** button. [[1](#), [2](#), [3](#)]
5. **Interact and Interrogate:**
- Look at the bottom left panel under **Deployed Contracts** to view the deployed instance.
 - Expand the contract block to expose the executable orange and blue buttons (storeNumber and getNumber). [[1](#), [2](#)]

Output

1. **Deployment Success:**

The Remix bottom console displays a green checkmark confirming successful transaction execution.

text

[vm] block: 1 | tx: 0x1a2b...3c4d | from: 0x5B3...eddC4 | to: SimpleStorage.(constructor)

Use code with caution.

[[1](#), [2](#), [3](#)]

2. **Executing storeNumber:**

- Type 88 inside the input box next to the orange storeNumber button, then click the button.

- **Console Log Output:**

text

status: true (Transaction mined successfully)

path: storeNumber

decoded input: { "uint256 _num": "88" }

logs: [

 { "topic": "NumberChanged", "args": ["88"] }

]

Use code with caution.

[[1](#)]

3. **Executing getNumber:**

- Click the blue getNumber button.

- **Console Log Output:**

text

decoded output: { "0": "uint256: 88" }

Use code with caution.

[[1](#)]



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Conclusion

The experiment successfully demonstrates writing, compiling, deploying, and invoking functions on a Solidity smart contract using Remix IDE. The application successfully altered and maintained its data state inside the transient blockchain sandbox environment. This confirms how decentralized ledger logic updates state variables through transaction calls while preserving data transparency using custom EVM log events.



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Experiment 3

Experiment: 3.Creating Transactions using Solidity and Remix IDE

Aim

To design, implement, and deploy an Ethereum smart contract in **Solidity** using **Remix IDE** to safely process and manage cryptocurrency transactions (sending, receiving, and checking balances).

Theory

1. Ether Transactions and Units

- **Value Transfer:** Smart contracts on Ethereum can natively hold, receive, and transfer **Ether** (ETH).
- **Denominations:** Solidity handles monetary values in **Wei**, the smallest unit of Ether (1 ETH = 10^{18} Wei). Other standard keywords include gwei (10^9 Wei). [[1](#), [2](#), [3](#), [4](#), [5](#)]

2. Specialized Solidity Modifiers & Functions

- **payable:** A required keyword for functions or addresses intended to receive incoming ETH. Without it, the transaction automatically reverts.
- **msg.value:** A global variable containing the exact amount of Wei sent along with the current transaction call.
- **msg.sender:** The Ethereum address that initiated the current transaction execution. [[1](#), [2](#), [3](#), [4](#), [5](#)]

3. Ether Transfer Mechanisms

Solidity provides three built-in primitive methods to send Ether from a contract to an external address: [[1](#)]

1. **transfer():** Forwards exactly 2300 gas; throws an error and reverts the transaction automatically on failure.
2. **send():** Forwards exactly 2300 gas; returns a boolean (true/false) status instead of reverting on failure.
3. **call():** Forwards all remaining transaction gas (unless specified otherwise); returns a boolean status and a data buffer. **This is the modern security standard** for transferring ETH. [[1](#), [2](#), [3](#), [4](#), [5](#)]



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Implementation / Procedure

Part 1: The Solidity Transaction Contract

Create a file named EtherTransaction.sol. This contract acts as an escrow system or money pool where any user can deposit funds, and only the owner can withdraw them to an external wallet address. [\[1\]](#)

solidity

```
// SPDX-License-Identifier: MIT
```

```
pragma solidity ^0.8.26;
```

```
contract EtherTransaction {
```

```
    // Address of the contract deployer (owner)
```

```
    address payable public owner;
```

```
    // Events to log transactions on the blockchain
```

```
    event DepositMade(address indexed sender, uint256 amount);
```

```
    event WithdrawalMade(address indexed receiver, uint256 amount);
```

```
    // Constructor sets the owner to the deployer's address
```

```
    constructor() {
```

```
        owner = payable(msg.sender);
```

```
    }
```

```
    /**
```

```
     * @dev Allows anyone to deposit Ether into this contract.
```

```
     * The 'payable' modifier enables the function to receive ETH.
```

```
     */
```

```
    function deposit() public payable {
```

```
        require(msg.value > 0, "You must send some Ether");
```

```
        emit DepositMade(msg.sender, msg.value);
```

```
    }
```

```
    /**
```

```
     * @dev Sends all contract-held funds back to the contract owner.
```

```
     */
```

```
    function withdrawAll() public {
```

```
        // Enforce access control: only the owner can call this
```

```
        require(msg.sender == owner, "Only the contract owner can withdraw funds");
```

```
        uint256 contractBalance = address(this).balance;
```

```
        require(contractBalance > 0, "No funds available to withdraw");
```

```
        // Use low-level call to safely transfer funds (Recommended Standard)
```

```
        (bool success, ) = owner.call{value: contractBalance}("");
```

```
        require(success, "Ether transfer failed");
```

```
        emit WithdrawalMade(owner, contractBalance);
```

```
    }
```



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```
/**
 * @dev Helper function to check the current balance of the contract.
 */
function getContractBalance() public view returns (uint256) {
    return address(this).balance;
}
}
```

Use code with caution.

Part 2: Step-by-Step IDE Procedure

1. **Open Remix:** Go to ethereum.org. [\[1\]](#), [\[2\]](#)
2. **Create File:** Navigate to the **File Explorer**, create a file named `EtherTransaction.sol`, and insert the code block above. [\[1\]](#)
3. **Compile Contract:** Click the **Solidity Compiler** sidebar icon, match your compiler version to 0.8.26, and click **Compile EtherTransaction.sol**. [\[1\]](#), [\[2\]](#), [\[3\]](#)
4. **Deploy Contract:**
 - Open the **Deploy & Run Transactions** tab.
 - Pick any account from the **Account** dropdown (e.g., Account 0x5B3... with 100 ETH).
 - Click **Deploy**. [\[1\]](#), [\[2\]](#)
5. **Simulate a Deposit:**
 - Change the interactive **Account** dropdown field to a *different* mock address (e.g., Account 0xA8b...) to act as a generic depositor.
 - Locate the **Value** input field above the deploy button. Change the unit dropdown from **Wei** to **Ether**, and input 5 in the box.
 - Scroll down to your deployed contract interface and click the red **deposit** button.
6. **Check Contract Balances:** Click the blue **getContractBalance** button.
7. **Simulate a Withdrawal:**
 - Switch the active **Account** dropdown field back to your original owner address (0x5B3...).
 - Click the orange **withdrawAll** button.

Output

1. Deployment Phase:

text

[vm] block: 1 | tx: 0x9e3c... | from: 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4

Use code with caution.

2. Deposit Execution (5 ETH from Account 0xA8b...):

- **Console Log Output:**

text

status: true (Transaction mined successfully)

value: 5000000000000000000 Wei (5 Ether)

path: deposit



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```
logs: [
  { "topic": "DepositMade", "args": ["0xAb84...", "50000000000000000000"] }
]
```

Use code with caution.

- **State Query (getContractBalance):**

Clicking the button prints: decoded output: { "0": "uint256: 50000000000000000000" } (5 ETH is securely safely locked inside the contract). [\[1\]](#)

3. Withdrawal Execution (Triggered by Owner 0x5B3...):

- **Console Log Output:**

text

status: true (Transaction mined successfully)

path: withdrawAll

```
logs: [
  { "topic": "WithdrawalMade", "args": ["0x5B38...", "50000000000000000000"] }
]
```

Use code with caution.

- **Result Verification:** Checking getContractBalance now returns 0. The balance of Account 0x5B3... jumps from its baseline back up to roughly 105 ETH (minus minor gas fees).

Conclusion

The experiment successfully demonstrates native asset handling inside Ethereum smart contracts using Solidity. By utilizing the payable function modifier and the state-standard .call{value: ...}("") syntax, transactional value transfers were handled deterministically. The execution confirmed how programmatic rules (such as require checks) prevent malicious state modification, maintaining strict ownership access controls over decentralized funds. [\[1\]](#)



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Experiment 4

Experiment: 4.Embedding Wallet and Transaction using Solidity

Aim

To design, write, deploy, and interact with a smart-contract-based **crypto wallet** embedded directly inside an Ethereum smart contract using **Solidity**. The wallet will be capable of securely receiving Ether, tracking internal user ledger balances via data structures, and executing arbitrary external transactions. [[1](#), [2](#), [3](#), [4](#), [5](#)]

Theory

1. Contract-Based Wallets vs. EOAs

An **Externally Owned Account (EOA)** is a user-controlled wallet managed via a private key (like a MetaMask account). An **Embedded Smart Contract Wallet** is an on-chain program that acts as a wallet. It can execute complex programmatic rules, enforce access controls, and automate transfers based on code conditions. [[1](#), [2](#), [3](#), [4](#), [5](#)]

2. Specialized Fallback Functions

To receive raw Ether transfers directly without forcing users to call a specific named function, a Solidity contract uses two special built-in handlers:

- **receive() external payable:** Triggered when the transaction data payload is completely empty (msg.data is empty).
- **fallback() external payable:** Triggered when a transaction sends Ether along with data specifying a function call that does not exist inside the contract interface. [[1](#), [2](#), [3](#), [4](#), [5](#)]

3. Solidity Mappings and Tracking

- **Nested Bookkeeping:** Instead of keeping track of just the global contract balance, an embedded multi-user wallet tracks individual account contributions using a dynamic hash table called a mapping: mapping(address => uint256). [[1](#)]
- **State Updates:** To prevent severe security issues like **Reentrancy attacks**, state variables (the mapping balances) must always be updated *before* executing the external transfer code block. [[1](#), [2](#)]



Implementation / Procedure

Part 1: The Embedded Wallet Smart Contract

Create a file named EmbeddedWallet.sol. This contract embeds custom wallet logic, tracks specific user deposits, and permits transactional routing. [[1](#), [2](#)]

solidity

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.26;
```

```
contract EmbeddedWallet {
    // Owner of the embedded wallet system
    address public walletOwner;

    // Mapping to track individual deposits embedded within the wallet ledger
    mapping(address => uint256) public userBalances;

    // Events to log embedded transactional operations
    event FundsReceived(address indexed depositor, uint256 amount);
    event TransactionExecuted(address indexed recipient, uint256 amount);
```

```
    modifier onlyOwner() {
        require(msg.sender == walletOwner, "Not the wallet owner");
        _;
    }
}
```

```
constructor() {
    walletOwner = msg.sender;
}
```

```
/**
 * @dev Automatic handler for direct raw incoming Ether transfers with no call data.
 */
```

```
receive() external payable {
    userBalances[msg.sender] += msg.value;
    emit FundsReceived(msg.sender, msg.value);
}
```

```
/**
 * @dev Named function allowing explicitly tracked wallet deposits.
 */
```

```
function depositToWallet() public payable {
    require(msg.value > 0, "Cannot deposit zero value");
    userBalances[msg.sender] += msg.value;
    emit FundsReceived(msg.sender, msg.value);
}
```

```
/**
```



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```
* @dev Initiates an external transaction out of the wallet to an arbitrary destination address.
* @param _to The destination address receiving the transaction.
* @param _amount The amount of funds in Wei to dispatch.
*/
function executeTransaction(address payable _to, uint256 _amount) public onlyOwner {
    require(_to != address(0), "Cannot send to zero address");
    require(address(this).balance >= _amount, "Insufficient wallet balance");

    // Execute the external native value transaction
    (bool success, ) = _to.call{value: _amount}("");
    require(success, "Transaction routing failed");

    emit TransactionExecuted(_to, _amount);
}

/**
* @dev Returns the total native asset value currently securely embedded inside the wallet.
*/
function getWalletTotalValue() public view returns (uint256) {
    return address(this).balance;
}
}
```

Use code with caution.

Part 2: Step-by-Step Execution Procedure

1. **Open Workspace:** Open Remix IDE. Create a file named EmbeddedWallet.sol and populate it with the source code above. [[1](#), [2](#)]
2. **Compile:** Go to the compiler interface, select version 0.8.26, and successfully compile the source code.
3. **Initial Deployment:**
 - Under **Deploy & Run Transactions**, use Account 1 (0x5B3...) as the administrator/owner.
 - Click the orange **Deploy** button. [[1](#)]
4. **Simulate a Direct Raw Asset Transfer:**
 - Switch the active workspace dropdown to Account 2 (0xAb8...).
 - Leave the *Function call buttons entirely alone*. Go straight to the **Value** block, type 3, and select **Ether**.
 - Scroll down to **Deploy & Run Transactions**, find the tiny **Low level interactions** panel near the bottom, and click the blank **Transact** button. This explicitly fires off the contract's implicit receive() function.
5. **Route an Outgoing Wallet Transaction:**
 - Switch back to the administrative controller account, Account 1 (0x5B3...).
 - Copy the raw hex address of an entirely fresh account, Account 3 (0x4B2...), to act as our external transaction receiver.
 - Expand the green **executeTransaction** block. Paste the copied address into the _to input slot.



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- Input the numerical transfer volume into the _amount slot in Wei units (e.g., 1000000000000000000 for 1 Ether).
- Click the transaction button. [[1](#), [2](#)]

Output

1. Raw Inbound Transaction Event Execution:

- **Action:** Account 2 sent 3 Ether directly to the contract address via low-level interactions.
- **Console Console Log:**
text
status: true (Transaction mined successfully)
value: 3000000000000000000 Wei
logs: [
 { "topic": "FundsReceived", "args": ["0xAb84...", "3000000000000000000"] }
]
Use code with caution.
- Checking userBalances("0xAb84...") returns 3000000000000000000.

2. Outgoing Wallet Transaction Dispatch:

- **Action:** Owner executed an external transaction forwarding 1 Ether out of the contract storage pool to Account 3 (0x4B2...).
- **Console Log Output:**
text
status: true (Transaction mined successfully)
path: executeTransaction
decoded input: { "address _to": "0x4B2...", "uint256 _amount": "1000000000000000000" }
logs: [
 { "topic": "TransactionExecuted", "args": ["0x4B2...", "1000000000000000000"] }
]
Use code with caution.
- **State Query Results:** Clicking getWalletTotalValue shows the total contract storage balance decreased from 3000000000000000000 down to 2000000000000000000 (2 Ether remaining). Account 3's native mock balance successfully increased by 1 Ether. [[1](#), [2](#), [3](#)]

Conclusion

The experiment successfully demonstrates the foundational logic needed to embed digital crypto wallet frameworks inside custom Solidity code. By combining low-level fallback functions (receive()), balance ledger mapping spaces, and low-level dynamic address call structures, the contract safely functioned as an intermediate ledger wallet. This programming methodology forms the core architectural basis for designing decentralized payment gateways, escrow applications, and automated treasury tracking systems.



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Experiment 5

Experiment: 5. Blockchain Platform Ethereum using Geth

Aim

To set up, configure, and initialize a local private blockchain network on the **Ethereum** platform using the Go-Ethereum (**Geth**) command-line client. [[1](#), [2](#), [3](#)]

Theory

1. Go-Ethereum (Geth) [[1](#), [2](#)]

- **Definition:** Geth is the official command-line interface and node implementation for the Ethereum protocol, written in Go.
- **Function:** It allows users to run a full Ethereum node, mine blocks, generate accounts, transfer funds, and deploy smart contracts locally or on live networks. [[1](#), [2](#), [3](#), [4](#), [5](#)]

2. The Genesis Block

- **Definition:** The genesis block is block number 0, the absolute structural starting point of any blockchain network.
- **Configuration:** A custom private network requires a genesis.json configuration file. This file establishes network parameters such as the **Chain ID** (to prevent transaction replay attacks), default mining difficulties, and pre-allocated account balances. [[1](#), [2](#), [3](#), [4](#), [5](#)]

3. Execution Layer Key Metrics

- **alloc:** Explicitly pre-funds specific developer wallet addresses with mock Ether before mining begins.
- **difficulty:** Determines how hard it is to discover a valid block hash. Setting this very low in a private environment ensures instant block generation during testing.
- **gasLimit:** The maximum amount of computational gas that can be consumed collectively by all transactions within a single block. [[1](#), [2](#), [3](#)]



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Implementation / Procedure

Step 1: Install Geth

Download and install the appropriate Geth package for your operating system (via Homebrew on macOS, apt repository on Ubuntu, or an executable installer on Windows). [[1](#), [2](#)]

Step 2: Establish the Custom Genesis Configuration [[1](#)]

Create a project workspace directory and save the following file layout exactly as genesis.json: [[1](#)]

```
json
{
  "config": {
    "chainId": 12345,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "istanbulBlock": 0
  },
  "difficulty": "0x1",
  "gasLimit": "0x8000000",
  "alloc": {
    "0x70997970C51812dc3A010C7d01b50e0d17dc79C8": {
      "balance": "1000000000000000000000000"
    }
  }
}
```

Use code with caution.

Step 3: Network Initialization

Open your terminal window inside your project workspace folder and initialize the private data directory using your configuration file:

```
bash
```

```
geth --datadir ./private_blockchain init genesis.json
```

Use code with caution.

Step 4: Launching the Active Node Console

Boot the newly instantiated private node infrastructure and open an active, interactive JavaScript runtime console directly within the runtime environment:

```
bash
```

```
geth --datadir ./private_blockchain --networkid 12345 --http --http.api "eth,net,web3,personal" console
```

Use code with caution.

Step 5: JavaScript Console Management Operations

Once inside the interactive Geth JavaScript console prompt (>), execute administrative commands to manage accounts and inspect the ledger: [[1](#)]



```
// Check the balance of the pre-funded address from the genesis file
eth.getBalance("0x70997970C51812dc3A010C7d01b50e0d17dc79C8")
```

```
// View the total list of tracked accounts
eth.accounts
Use code with caution.
```

1. Initialization Phase Terminal Output:

```
$ geth --datadir ./private_blockchain init genesis.json
INFO [07-23|15:42:01.102] Maximum peer count                ETH=50 LES=0 total=50
INFO [07-23|15:42:01.115] Allocated cache and file handles
database=./private_blockchain/geth/chaindata cache=16.00MiB handles=16
INFO [07-23|15:42:01.150] Writing custom genesis block
INFO [07-23|15:42:01.162] Successfully wrote genesis state   database=chaindata hash=5e1d3c...
Use code with caution.
```

text

```
instance: Geth/v1.14.0-stable/darwin-arm64/go1.22.1
modules: admin:1.0 debug:1.0 eth:1.0 miner:1.0 net:1.0 personal:1.0 txpool:1.0 web3:1.0
```

```
> eth.getBalance("0x70997970C51812dc3A010C7d01b50e0d17dc79C8")  
1000000000000000000
```

```
> personal.newAccount("securePassword123")
"0x3c4ax9b...a12c45f"
```

```
> eth.blockNumber
0
```

Use code with caution.

[1, 2]



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Conclusion

The experiment successfully demonstrates how to configure and deploy a fully functional standalone, private Ethereum blockchain layer using the **Geth** client software. By writing a custom genesis.json profile, we established a localized consensus environment with custom gas thresholds and initial developer balances. This localized sandbox runtime environment allows developers to test decentralized apps and transaction structures deterministically without paying live gas fees.



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Experiment 6

Experiment: 6.Blockchain Platform using Ganache

Aim

To set up, configure, and interact with a rapid, local development Ethereum blockchain network using the **Ganache** platform. [[1](#), [2](#)]

Theory

1. What is Ganache?

- **Definition:** Ganache is a personal, localized, zero-config blockchain simulator provided by the Truffle/Consensus suite.
- **Purpose:** It is designed specifically for local smart contract development and testing, serving as a faster, more practical alternative to running a full-scale heavy node client like Geth. [[1](#), [2](#), [3](#), [4](#), [5](#)]

2. Key Architectural Features

- **Instant Mining (Zero Latency):** Unlike public networks or local Geth testnets that require proof-of-work or proof-of-authority intervals, Ganache processes and mines transactions instantly into blocks with zero wait time. [[1](#), [2](#), [3](#)]
- **Deterministic Sandboxing:** By default, Ganache generates **10 mock Ethereum accounts**, each pre-funded with **100 test Ether (ETH)**. These accounts are derived from a standard 12-word mnemonic phrase, meaning the same addresses can be regenerated deterministically. [[1](#), [2](#), [3](#)]
- **Built-in Logging and RPC:** It exposes an active JSON-RPC HTTP server endpoint (usually <http://127.0.0.1:7545> or <http://127.0.0.1:8545>). This endpoint handles requests from Web3 applications, Remix IDE, Hardhat, or MetaMask. [[1](#), [2](#), [3](#), [4](#)]

Implementation / Procedure

Step 1: Installation

Ganache can be run in two ways:



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1. **Ganache UI (Graphical Interface):** Download and run the standalone desktop installer from the official repository or website.
2. **Ganache CLI (Command Line):** Open a terminal and install it globally using Node Package Manager (NPM):

```
bash
```

```
npm install -g ganache
```

```
Use code with caution.
```

```
[1, 2, 3, 4, 5]
```

Step 2: Launching the Private Network

- **Using UI:** Open the application and click **Quickstart**.
- **Using CLI:** Launch the blockchain simulator by running the following command in your terminal:

```
bash
```

```
ganache
```

```
Use code with caution.
```

```
[1, 2, 3, 4, 5]
```

Step 3: Connecting Remix IDE to the Ganache Network

1. Open the Remix IDE web browser platform.
2. Create or load a basic smart contract (e.g., the SimpleStorage.sol contract built in previous sessions).
3. Compile the contract successfully.
4. Navigate to the **Deploy & Run Transactions** sidebar tab.
5. Click the **Environment** dropdown menu and switch from *Remix VM* to **Dev - Ganache Provider** (or select *External HTTP Provider*).
6. Input the target Ganache network endpoint URI when prompted: `http://127.0.0.1:8545` (CLI) or `http://127.0.0.1:7545` (UI). [1, 2, 3, 4, 5]

Step 4: Interacting and Executing State Changes

1. Observe that the Remix **Account** selection dropdown is now automatically populated with the 10 pre-funded accounts originating directly from Ganache.
2. Select the first address (0x90F...) and click **Deploy**.
3. Execute a state transaction (e.g., store a value or trigger a mock balance transfer). [1]

Output

1. **Ganache Core Startup Environment Log (CLI):**

```
text
```

```
$ ganache
```

```
ganache v7.9.2 (@ganache/cli: v0.10.2, @ganache/core: v0.10.2)
```

```
Starting RPC server
```

```
Available Accounts
```

```
=====
```

```
(0) 0x90F11C364C9b48E9d04B29c395e6922154Cb14b4 (100 ETH)
```

```
(1) 0x3C4Ax9b2A4d12c45f470997970C51812dc3A010C (100 ETH)
```



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...

Private Keys

=====

(0) 0x4f3edf983ac636a65a842ce7c78d9aa706d3b113bce9c46f30d7d21715b23b1d

...

HD Wallet

=====

Mnemonic: candy maple cake sugar pudding cream honey rich smooth crumble sweet treat

Base HD Path: m/44'/60'/0'/0'/account_index

RPC Listening on 127.0.0.1:8545

Use code with caution.

2. **Transaction Settlement Log (Triggered via Remix Interface):**

When you deploy your smart contract through the Remix endpoint, the Ganache server console streams structural payload transaction logs in real time:

text

eth_sendTransaction

Transaction: 0x2d1a3c...7e4b

Contract Created: 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4

Gas Used: 124312 of 6721975

Block Number: 1

Time: Thu Jul 23 2026 15:43:00 GMT

eth_getTransactionReceipt

eth_blockNumber

Use code with caution.

*In Ganache UI, these details are visually rendered inside the **Transactions** and **Blocks** navigation tabs. [\[1\]](#)*

Conclusion

The experiment successfully demonstrates using **Ganache** as a lightweight, high-speed local simulation platform for Ethereum networks. By binding Ganache's local RPC listener endpoint directly to Remix IDE, we eliminated block mining latency entirely. This setup provides developer environments with a fast, safe, and cost-free sandbox to observe real-time transaction processing, gas usage, and account balance changes before moving to a live testnet



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Experiment 7

Experiment: 7. Case Study on Hyperledger

1. Title & Abstract

- **Title:** Resolving Global Logistics Friction and Fraud via Permissioned Distributed Ledgers: A Case Study on Hyperledger Fabric.
- **Abstract:** This study examines how enterprise consortia leverage Hyperledger Fabric's permissioned blockchain architecture to eliminate paper-based tracking silos, secure trade finance pipelines, and automate regulatory compliance while enforcing strict commercial data privacy. [\[1\]](#)

2. The Business Problem & Challenges

Global supply chains historically suffer from systemic inefficiencies due to fragmented operating environments:

- **Information Silos:** Carriers, ports, customs brokers, and banks track cargo on separate, incompatible internal databases.
- **Documentation Fraud:** Paper bills of lading are easily forged, altered, or duplicated, leading to billions in annual trade finance losses.
- **Data Privacy Concerns:** Competitors hesitate to use open public blockchains (like Ethereum) because public ledgers expose transaction volumes, pricing structures, and counterparty identities. [\[1\]](#)
- **Performance Constraints:** Public networks incur volatile transaction (gas) fees and slow throughput speeds that cannot support high-frequency logistical tracking.

3. Why Hyperledger Fabric? (The Solution Architecture)

Hyperledger Fabric was chosen over public networks due to key architectural features engineered specifically for enterprise business rules: [\[1, 2, 3\]](#)

A. Permissioned Access Control [\[1\]](#)

- Every participant must verify their identity using standard cryptographic certificates.



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- Access is managed via a **Membership Service Provider (MSP)**, keeping bad actors out of the network. [1, 2, 3, 4]

B. Data Isolation via Private Channels [1, 2]

- Fabric allows users to spin up isolated communication networks called **Channels**.
- For example, a manufacturer can negotiate private cargo pricing with a shipping line over a specific channel, ensuring competing shipping networks on the broader ledger cannot see the financial terms. [1, 2, 3]

C. High Performance & Low Cost

- The network eliminates gas fees entirely.
- It replaces expensive proof-of-work mining with deterministic consensus algorithms (like **Raft**), enabling the platform to process over **3,000+ transactions per second (TPS)** with sub-second finality. [1]

4. Technical Implementation & Workflow

[Exporter Node] —(1. Endorse)—> [Endorsing Peers] —(2. Validated Signature)

(4. Append to Ledger) <— [Orderer Node Cluster] <—(3. Broadcast Transact Payload)

The case study architecture follows an **Execute-Order-Validate** workflow, which is unique to Hyperledger Fabric: [1, 2, 3]

1. **Transaction Proposal (Execute):** The exporter generates a digital shipping milestone document. The asset status change is previewed and cryptographically signed (endorsed) by designated peer nodes running smart contracts (**Chaincode**). [1, 2]
2. **Ordering Service (Order):** Endorsed transactions are collected and broadcasted to an independent **Orderer Node Cluster**. The cluster organizes these transactions chronologically into raw data blocks without risking fork splits. [1, 2, 3]
3. **Validation & Commitment (Validate):** The finalized blocks are distributed to all authorized peer nodes on the channel. Each node verifies the cryptographic signatures before permanently committing the block data to its local state database (**CouchDB**). [1, 2]

5. Measurable Business Outcomes & Metrics

Real-world enterprise network rollouts (such as IBM's Food Trust and various global maritime tracking platforms) demonstrate clear efficiency gains:

- **Administrative Speed:** Cross-border paperwork verification times dropped from **average windows of 7-10 days down to less than 2 minutes**.
- **Financing Risk Mitigation:** Importers can instantly verify genuine bills of lading on-chain, **reducing trade finance fraud attempts to near 0%**.



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- **Cold-Chain Freshness:** Integrated IoT sensors stream ambient temperature data to the ledger every 15 minutes, cutting food spoilage waste by **up to 40%**.

6. Critical Evaluation & Challenges

- **Consensus Governance:** Convincing dozens of international shipping lines and competing financial institutions to agree on a shared network configuration remains a complex legal and operational challenge.
- **Infrastructure Complexity:** Running specialized, on-premises or cloud-hosted peer nodes requires an experienced engineering team, resulting in higher initial setup costs compared to simple cloud services.

7. Case Study Conclusion

Hyperledger Fabric demonstrates that permissioned blockchains are highly effective at solving enterprise data fragmentation. By combining strict identity verification with isolated private data channels, Fabric bridges the gap between trust and data confidentiality. The platform allows modern enterprise consortia to replace slow, paper-heavy workflows with secure, automated, and tamper-proof asset ledger systems.



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Experiment 8

Experiment: 8. Case Study on Other Blockchain Platforms

Case Study 1: Ripple (XRP Ledger) for Cross-Border Settlement

A. The Business Problem

Traditional international bank transfers rely on the legacy **SWIFT** network and a complex system of **Nostro/Vostro correspondent banking accounts**. This system suffers from high costs, long delays (3–5 business days), and high transaction failure rates due to manual processing. [[1](#), [2](#), [3](#), [4](#), [5](#)]

C. The Blockchain Architecture Solution

D.

Financial institutions leverage the **XRP Ledger (XRPL)** and Ripple's payment solution to conduct high-speed settlement. [[1](#), [2](#), [3](#)]

- **Consensus Mechanism:** Uses the Ripple Protocol Consensus Algorithm (RPCA) rather than proof-of-work. A network of independent validating nodes constantly checks transaction sets. [[1](#), [2](#), [3](#)]
- **On-Demand Liquidity (ODL):** Uses **XRP** as a digital bridge asset. A bank converts local fiat currency (e.g., USD) to XRP, sends it across the ledger in seconds, and converts it instantly to the destination fiat currency (e.g., EUR) on the receiving side. [[1](#), [2](#), [3](#), [4](#), [5](#)]

C. Real-World Outcomes

- **Settlement Speeds:** Transaction settlement times plummeted from **3-5 business days to roughly 3-5 seconds**.
- **Cost Efficiency:** Avoids trapped capital in foreign Nostro accounts, **slashing payment processing fees by up to 40-70%**. [[1](#), [2](#), [3](#), [4](#), [5](#)]

Case Study 2: Corda (R3) for Private Financial Agreements

A. The Business Problem

In fields like derivatives trading, syndication, and reinsurance, corporations must share complex legal contracts without leaking data. While Hyperledger Fabric uses channels to isolate data, channels can become bloated and complex when thousands of distinct, bilateral agreements require individual isolation. [[1](#), [2](#)]



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B. The Blockchain Architecture Solution

Corda (developed by R3) is an open-source DLT built specifically for highly regulated financial financial systems. [[1](#), [2](#), [3](#), [4](#)]

- **No Global Broadcasts:** Unlike traditional blockchains that broadcast data to all nodes, Corda only shares data on a **strict need-to-know basis**. If Bank A transactions with Bank B, only those two nodes store the data. [[1](#), [2](#)]
- **Notary Clusters:** Uses specialized "Notary" nodes to prevent double-spending and guarantee transaction finality without exposing the transaction details to the rest of the network. [[1](#), [2](#), [3](#), [4](#)]
- **Legal Grounding:** Designed so smart contract code is explicitly tied to legal prose, ensuring agreements remain enforceable in traditional courts. [[1](#), [2](#)]

D. Real-World Outcomes

- **Absolute Privacy:** Competitors operate on the exact same network without ever seeing each other's activity, transaction volumes, or data footprints. [[1](#)]
- **Regulatory Compliance:** Built-in features seamlessly accommodate KYC/AML laws and data privacy mandates like GDPR, as data is never broadcasted to an open ledger. [[1](#), [2](#), [3](#), [4](#)]

Case Study 3: Polkadot for Cross-Chain Healthcare Interoperability

A. The Business Problem

Medical records, pharmacy supply metrics, and clinical trial results are often locked inside separate, isolated public and private blockchains. This isolation makes it incredibly difficult to securely transfer data across different networks without relying on risky centralized bridges. [[1](#), [2](#), [3](#)]

B. The Blockchain Architecture Solution

Polkadot functions as a sharded multi-chain meta-protocol, acting as a foundation layer that connects separate blockchains. [[1](#), [2](#)]

- **The Relay Chain:** The heart of the network. It handles network security, consensus, and cross-chain interoperability for all connected chains. [[1](#), [2](#), [3](#), [4](#), [5](#)]
- **Parachains:** Independent, custom-built blockchains designed for specific use cases (e.g., a private healthcare data blockchain) that plug directly into the Relay Chain. [[1](#), [2](#), [3](#), [4](#), [5](#)]
- **XCM (Cross-Consensus Messaging):** A shared communication format that allows connected chains to securely exchange medical records or supply tokens without using vulnerable third-party bridges. [[1](#), [2](#)]



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C. Real-World Outcomes

- **Shared Security:** Smaller medical blockchains don't need to build their own expensive validator networks; they inherit top-tier security directly from Polkadot's central Relay Chain.
- **True Interoperability:** A private patient tracking blockchain can seamlessly trigger an insurance payout on a public financial blockchain securely and instantly. [[1](#), [2](#)]

Key Takeaway Comparison

Feature	Ripple (XRPL)	Corda (R3)	Polkadot
Primary Focus	Global Payments	Financial Contracts	Multi-Chain Interoperability
Data Sharing	Public Ledger	Point-to-Point (Private)	Shared Across Connected Chains
Native Token	Required (XRP)	No Native Token	Required (DOT)
Best Used For	Cross-border remittance	Regulated banking & trade	Ecosystem cross-communication



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Experiment 9

Experiment: Creating a Blockchain Application

Aim

To design, implement, and run a decentralized **Full-Stack Blockchain Application** (DApp) that allows users to interact with a smart contract using a web browser interface via MetaMask. [1]

Theory

1. Decentralized Application (DApp) Architecture [1]

A typical DApp separates responsibilities across three distinct architectural tiers: [1]

- **Frontend UI:** Built with standard web technologies (HTML, JavaScript, React). It captures user inputs and visualizes blockchain data.
- **Web3 Provider/Wallet (MetaMask):** Acts as a secure identity bridge. It securely holds the user's private keys, signs transactions, and maintains an active JSON-RPC connection to the blockchain.
- **Smart Contract Layer:** The backend business logic deployed immutably to a blockchain execution environment (EVM). [1, 2, 3, 4, 5]

[Frontend UI] —(Interactions via Ethers.js)—> [MetaMask Wallet] —(Signs Tx)—> [Blockchain Network]

2. Ethers.js / Web3.js Libraries

Frontend web browsers cannot communicate with blockchain nodes using raw JavaScript. Standard JavaScript applications rely on library wrappers like **Ethers.js** or **Web3.js** to parse smart contract **ABIs (Application Binary Interface)** and format standard client requests into structured JSON-RPC payloads. [1, 2, 3, 4]

Implementation / Procedure

Part 1: The Smart Contract Backend (MessageBoard.sol)

Create a basic smart contract that tracks a globally modifiable message string and emits block events when updated. [1, 2]

solidity



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```
// SPDX-License-Identifier: MIT
```

```
pragma solidity ^0.8.26;
```

```
contract MessageBoard {
```

```
    string public currentMessage;
```

```
    event MessageUpdated(address indexed user, string newMessage);
```

```
    constructor() {
```

```
        currentMessage = "Welcome to the Blockchain Application!";
```

```
    }
```

```
    function updateMessage(string memory _newMessage) public {
```

```
        require(bytes(_newMessage).length > 0, "Message cannot be empty");
```

```
        currentMessage = _newMessage;
```

```
        emit MessageUpdated(msg.sender, _newMessage);
```

```
    }
```

```
}
```

Use code with caution.

- **Procedure:** Deploy this contract using **Remix IDE** or **Hardhat** to a local development network (like Ganache or Remix VM) and copy its deployed **Contract Address** and **ABI array**. [[1](#), [2](#), [3](#), [4](#), [5](#)]

Part 2: The Frontend UI Framework (index.html) [[1](#), [2](#), [3](#)]

Create an HTML application that connects to MetaMask and interacts with your deployed contract using the Ethers.js library. [[1](#), [2](#), [3](#), [4](#)]

html

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
    <meta charset="UTF-8">
```

```
    <title>Blockchain DApp</title>
```

```
    <!-- Include Ethers.js via secure CDN -->
```

```
    <script src="https://cloudflare.com" type="text/javascript"></script>
```

```
    <style>
```

```
        body { font-family: Arial, sans-serif; padding: 30px; text-align: center; background: #f4f6f9; }
```

```
        .card { background: white; padding: 25px; border-radius: 8px; box-shadow: 0 4px 6px rgba(0,0,0,0.1);
```

```
display: inline-block; max-width: 400px; }
```

```
        button { background: #007bff; color: white; border: none; padding: 10px 20px; border-radius: 4px; cursor: pointer; font-size: 14px; margin-top: 10px; }
```

```
        input { padding: 10px; width: 80%; margin-top: 10px; border: 1px solid #ccc; border-radius: 4px; }
```

```
    </style>
```

```
</head>
```

```
<body>
```

```
<div class="card">
```

```
    <h2>Blockchain DApp</h2>
```

```
    <button id="connectBtn" onclick="connectWallet()">Connect MetaMask Wallet</button>
```

```
    <p id="walletAddress" style="color: green; font-weight: bold;"></p>
```

```
</div>
```



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```
<h3>Current On-Chain Message:</h3>
<p id="displayMessage" style="font-style: italic; color: #555;">Loading...</p>
<button onclick="readMessage()">Refresh Message</button>

<hr/>

<input type="text" id="newMessageInput" placeholder="Enter a new message..." />
<button onclick="writeMessage()">Submit On-Chain Transaction</button>
</div>

<script>
  // Configuration Variables (Replace with your actual deployment artifacts)
  const contractAddress = "0x5B38Da6a701c568545dCfcB03FcB875f56beddC4";
  const contractABI = [
    "function currentMessage() view returns (string)",
    "function updateMessage(string memory _newMessage) public"
  ];

  let provider, signer, contract;

  async function connectWallet() {
    if (typeof window.ethereum !== 'undefined') {
      try {
        // Request account access from MetaMask
        const accounts = await window.ethereum.request({ method: 'eth_requestAccounts' });
        document.getElementById("walletAddress").innerText = `Connected:
${accounts[0].substring(0,6)}...${accounts[0].substring(38)}`;

        // Initialize Ethers provider and signer
        provider = new ethers.providers.Web3Provider(window.ethereum);
        signer = provider.getSigner();
        contract = new ethers.Contract(contractAddress, contractABI, signer);

        document.getElementById("connectBtn").style.display = "none";
        readMessage();
      } catch (error) {
        console.error("Wallet connection rejected", error);
      }
    } else {
      alert("Please install MetaMask to interact with this application!");
    }
  }

  async function readMessage() {
    if (!contract) return;
    try {
      const msg = await contract.currentMessage();
      document.getElementById("displayMessage").innerText = `"${msg}"`;
    } catch (err) {
      console.error("Error reading contract data:", err);
    }
  }
</script>
```



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```
}  
}  
  
async function writeMessage() {  
  if (!contract) return alert("Please connect your wallet first!");  
  const inputVal = document.getElementById("newMessageInput").value;  
  if (!inputVal) return alert("Message cannot be empty!");  
  
  try {  
    // Trigger smart contract write transaction  
    const tx = await contract.updateMessage(inputVal);  
    document.getElementById("displayMessage").innerText = "Processing transaction in block...";  
  
    // Wait for transaction confirmation on-chain  
    await tx.wait();  
  
    document.getElementById("newMessageInput").value = "";  
    readMessage();  
  } catch (err) {  
    console.error("Transaction failed or canceled:", err);  
  }  
}  
</script>  
</body>  
</html>  
Use code with caution.
```

Output

1. **Wallet Initialization:**

Clicking the **Connect MetaMask Wallet** button opens the MetaMask browser extension window. Once confirmed, the button disappears, and the shortened user address is displayed:

Connected: 0x5B38...eddC4

Current On-Chain Message: "Welcome to the Blockchain Application!"

Use code with caution.

[\[1\]](#)

2. **State Modification Execution:**

- The user enters Hello World Blockchain! into the input field and clicks **Submit On-Chain Transaction**.
- MetaMask prompts the user to confirm the gas fee transaction signature.
- **Console Event Processing Sequence:**

text

[DApp Context] Broadcast transaction: 0x4f8c...a21b

[DApp Context] Waiting for block mining confirmation...

[DApp Context] Transaction confirmed in Block #14. Fetching updated state...

Use code with caution.



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- The web page automatically updates to display the new text fetched from the contract storage:

text

Current On-Chain Message: "Hello World Blockchain!"

Use code with caution.

[\[1\]](#), [\[2\]](#), [\[3\]](#)

Conclusion

The experiment successfully demonstrates how to build and run an end-to-end full-stack decentralized application (DApp). By combining a Solidity smart contract backend with an HTML5/Ethers.js frontend user interface, we eliminated the need for traditional centralized database management components. Instead, state variables are read directly from and securely updated on the blockchain network using cryptographic transaction signatures authorized via MetaMask.